# Poojan Shah

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## Research Interests

My current focus is the optimization dynamics of pretraining large-scale transformers, with the goal of recipes that are more efficient, stable and predictable as models scale. Earlier work on clustering algorithms and beyond-worst-case algorithm design.

## Employment

August 2026–Present — AI Scientist, Pretraining Team, Mistral AI

## Education

2026 — **B.Tech. in Computer Science and Engineering, Indian Institute of Technology Delhi**
GPA: 9.75/10.00 (Department Rank 3) Advisor: Prof. Ragesh Jaiswal

2025 — **Visiting Undergraduate Student, Faculty of Mathematics, University of Waterloo**
Grade: 94.20/100.00

## Internships

Nov 2025–Feb 2026 — Research Intern, Wadhwani AI
Host: Makarand Tapaswi
*Self-supervised learning from unlabeled data for newborn anthropometry*

May 2025–July 2025 — Quantitative Research Intern, Atlas Research
*Optimized exchange data ingestion and API calls; developed statistical price-forecasting models and parameter-search algorithms*

May 2024–July 2024 — Undergraduate Research Intern, Centre for Quantum Technologies, NUS
Host: Rahul Jain
*Studied cryptographic primitives based on pseudorandom quantum states*

## Selected Publications

1. **Noisy $k$-means++ Is Not *Too* Noisy**
*Poojan Shah*. Preprint, 2026 [arXiv]

2. **Fast $k$-means Seeding Under the Manifold Hypothesis**
*Poojan Shah*, *Shashwat Agrawal*, *Ragesh Jaiswal*. ICML 2026 [arXiv] [OpenReview] [Poster]

3. **Quantum (Inspired) $D^2$-Sampling with Applications**
*Poojan Shah*, *Ragesh Jaiswal*. ICLR 2025 [arXiv] [OpenReview] [Poster]

## Selected Awards & Honors

1. Suresh Chandra Memorial Trust Award *(best B.Tech. Thesis project in the CSE Dept.)*
2. Amit and Deepali Sinha Foundation Fellowship, 2022
*Four-year full-ride undergraduate scholarship; top five at IIT Delhi by JEE Advanced rank*
3. CSE Research Acceleration Award *(funding for research and conference travel)*, 2025–2026
4. Selected Participant, Cornell–Maryland–Max Planck Pre-doctoral Research School, 2025
5. All India Rank 67, JEE Advanced, 2022

## Talks

**Quantum and Quantum-Inspired Classical Algorithms for Clustering**
Feb 2026 — Invited Lightning Talk, ACM India ARCS 2026, IIT Hyderabad
Apr 2025 — CS Group Meeting, Centre for Quantum Technologies, NUS [Poster]

**Quantum Machine Learning without any Quantum**
Nov 2024 — TCS Seminar, IIT Delhi [Slides]

## Teaching

Winter 2026 — **Head Teaching Assistant, COL7160: Quantum Computing**
IIT Delhi Recipient of the CSE Department's Outstanding Teaching Assistant Award.